

Lecture 38

STAT 109: Introductory Biostatistics - Ethics and statistical reasoning review

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Data as evidence, analyst as a truth-first advocate

In this course, think of **data as evidence**. As analysts, we are not hired to defend one side at any cost; we are responsible for interpreting evidence as honestly as possible.

That means:

- we do not cherry-pick results,
- we do not overstate certainty,
- we do not hide missingness or failed assumptions,
- we do not make claims stronger than design supports.

Statistics and ethics are inseparable.

Step 1: How was the evidence generated?

- **Random sample (SRS):** supports stronger population generalization.
- **Convenience sample:** often fast, but likely biased with limited representativeness.
- **Random assignment (RCT):** supports stronger causal interpretation.
- **Observational study:** often more feasible than an RCT, but there is a risk of seeing a difference that is entirely due to confounding.

Question: *What claims are justified by this design, and what claims are not?*

Step 2: What do we see descriptively?

- Use appropriate plots/summaries for variable types.
- Describe form, direction, strength, and group differences carefully.
- Keep descriptive and inferential language separate.

Questions: Is the type of plot I chose appropriate for the type of data I have? Do choices like bin-width in a histogram affect my interpretation? Am I remembering to be careful to consider the aspects of the data that I can't see in a plot (such as sampling method and study design) when making a conclusion?

Step 3: What does inference add?

- Use valid methods for the variable types and design.
- Interpret p-values/CIs in context and with uncertainty.
- Check assumptions/conditions before trusting p-values and intervals.

Question: *Are our mathematical conclusions valid under these data conditions?*

IRB and duty to participants

For human-subject research, Institutional Review Board (IRB) oversight protects people and the integrity of evidence.

Core protections include:

- minimizing harm,
- informed consent,
- fair participant selection,
- privacy and confidentiality,
- risk/benefit justification.

If randomization would deliberately expose people to clear harm (for example, assigning smoking initiation), that design is unethical even if it is statistically powerful.

Professional reference: American Statistical Association, *Ethical Guidelines for Statistical Practice*
<https://www.amstat.org/your-career/ethical-guidelines-for-statistical-practice>

Real examples: when ethics failures break inference

1) Coercive or non-consensual research

Some historical studies enrolled vulnerable groups without modern informed-consent protections.

- Ethical failure: autonomy and respect for persons violated and our evidence is contaminated by coercion and harm.
- Lesson: participant protection is part of data quality.

2) Harmful assignment proposals

Proposal: randomize teens to start smoking vs not smoking.

- Ethical failure: intentional harm.
- Statistical failure: “clean design” cannot justify unethical treatment.
- Lesson: some causal questions must be studied observationally.

3) Convenience sample overclaim

Volunteers from one platform are used to claim national conclusions.

- Ethical failure: misleading communication because of unknown selection bias, weak external validity.
- Lesson: generalize only as far as sampling supports.

4) Causal headline from observational data

Single observational association reported as “X causes Y.”

- Ethical failure: overstating evidence.
- Statistical failure: confounding/selection bias/chance ignored.
- Lesson: weak language is required unless causal design supports stronger claim.

5) Hidden attrition

A treatment report excludes participants who dropped out from side effects.

- Ethical failure: selective reporting means that effect estimates may be biased upward.
- Lesson: dropout/missingness must be reported and discussed.

Missing responses are evidence too

Missing data are not just blank cells; they can reveal important bias.

If dropout is related to treatment or outcome:

- complete case analysis where only rows with full information are used can distort effects,

- uncertainty can be understated, and thus
- conclusions can become misleading.

Minimum truth-first reporting:

- counts missing by group,
 - reasons for missing/dropout when known,
 - comparison of completers vs non-completers when feasible,
 - sensitivity discussion: how could missingness change conclusions?
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Ethical interpretation rules by topic

Generalization

- If sample is convenience-based and so likely biased, say so directly.
- Do not frame sample findings as population truth.

Causation

- RCTs can support stronger causal claims (with caveats).
- Observational studies require confounding alternative in conclusions.

Statistical significance

- “Small p-value” does not mean “proved true.”
 - “Fail to reject” does not prove no effect.
 - Remember the p in p-value stands for probability! There is also the chance we are making a type I or type II error when conducting a test with a p-value.
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Scenario set

A) Convenience sample claim

A wellness app surveys Monday-morning users and claims “all college students.”

Discuss:

1. What can be claimed honestly?
2. What cannot be generalized?
3. How should conclusion wording change?

B) Observational causation claim

An observational stress study is released as “Meditation causes lower stress.”

Discuss:

1. Plausible alternative explanations.
2. Ethical wording of conclusions.
3. What stronger design would be needed for causal confidence?

C) Unethical randomization

A proposal suggests randomizing smoking initiation.

Discuss:

1. Why IRB should reject this.
2. What observational alternative could still produce useful evidence?

D) Nonreported dropout

A trial omits differential attrition from the main report.

Discuss:

1. How effect estimates could be biased.
2. Minimum reporting needed for ethical interpretation.

Language rewrites: from advocacy to truth

- Instead of: “We proved treatment causes improvement.”
Use: “In this randomized study, results are consistent with a treatment effect, with uncertainty summarized by CI/p-value.”
 - Instead of: “These volunteers represent everyone.”
Use: “Because this is a convenience sample, conclusions are limited to similar participants.”
 - Instead of: “Missing values were dropped.”
Use: “We report missingness/dropout patterns and discuss potential impact on conclusions.”
 - Instead of: “ $p < 0.05$, therefore true.”
Use: “Data are inconsistent with the null under model assumptions; conclusions remain probabilistic.”
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Summary

1. **Design honesty:** Are claims matched to sampling and assignment?
2. **Data integrity:** Is missingness/dropout transparently reported?
3. **Method validity:** Were assumptions checked and disclosed?
4. **Uncertainty clarity:** Are p-values/CIs interpreted correctly?
5. **Communication ethics:** Is wording accurate, limited, and non-misleading?